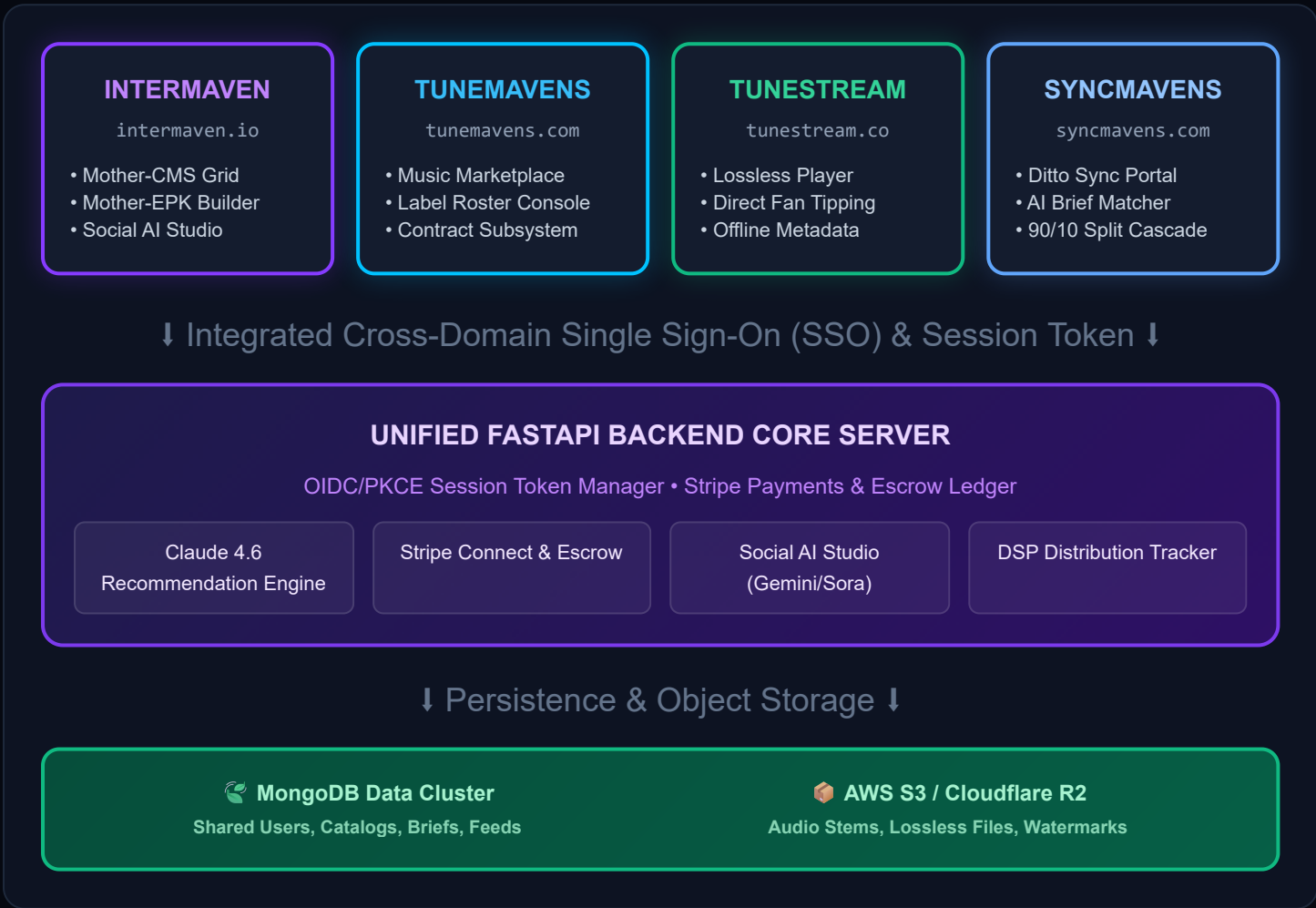


ECOSYSTEM SYSTEM ARCHITECTURE & FLOWCHART

Monorepo Technical Blueprint, Connectivity Flowchart & Data Schema Index

1. High-Impact Ecosystem Connectivity Flowchart



2. End-to-End Data & Asset Lifecycle Sequence

- Catalog Ingestion:** Creator uploads audio stems to TuneMavens. Stems store in AWS S3, metadata upserts to MongoDB catalogs with auto-generated ISRCs.
- Split Cascade Lock:** Creator defines split waterfall (90/10) in `publishing_deals` and signs contract via `contracts`.
- Sync Placement:** Music supervisor posts brief on SyncMavens. AI Match Simulator evaluates track compatibility. Pitch delivered with watermarked preview. Supervisor approves and 90/10 royalty cascade executes.
- Consumer Streaming:** Track publishes to TuneStream. Fans stream lossless audio & tip artist directly. Revenue credits route to unified network balance.
- Social AI Marketing:** Creator triggers Intermaven Social AI Studio to render 1:1 and 9:16 creative assets (Gemini/Sora) and dispatch via Resend email / social DM.

3. Data Schema & MongoDB Collection Index

Collection Name	Key Fields & Attributes	Primary Consuming Apps	Business Purpose
users	_id, email, roles[], primary_role, plan, credits	All 4 Platforms	Unified SSO identity, role permissions, and credit balance
catalogs	_id, isrc, title, stems_s3_url, splits[]	TuneMavens, SyncMavens, TuneStream	Centralized metadata index for masters, stems, and audio assets
briefs	_id, supervisor_id, budget, deadline, genre	SyncMavens	Licensing opportunities submitted by music supervisors
pitches	_id, brief_id, catalog_id, match_score, waterfall_splits	SyncMavens, TuneMavens	Pitch submissions and licensing deal ledgers
publishing_deals	_id, user_id, tier, splits_cascade, status	TuneMavens	Publishing agreement records and split calculation rules
contracts	_id, contract_type, locked_clauses[], signatures[]	TuneMavens, SyncMavens	E-signature and legal contract negotiation state machine
campaigns	_id, name, segment, channels[], stats	Intermaven, TuneMavens	Multi-channel CRM growth marketing campaigns